Chapter 2. Matrix, Calculus and Probability Tools

Matrix identities, eigenvalue inequalities, matrix calculus, optimization, convergence

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This chapter collects the tools used in later proofs (MLE, Wishart, T^2, large-sample theory).

1 Basic Matrix Operations

Motivation. This chapter collects the linear-algebra and probability tools used throughout, so later proofs can cite them. We start with trace, determinant, inverses of partitioned matrices and Kronecker products.

Definition 2.1: basic facts

(i) $(\mathbf{AB})_{ik} = \sum_j a_{ij}b_{jk}$; $(\mathbf{A}^\top)^\top = \mathbf{A}$, $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$; a 1×1 matrix equals its transpose.

(ii) $\operatorname{tr}(\mathbf{C}) = \sum_i c_{ii}$ is linear.

(iii) $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$, so $|\mathbf{A}^{-1}| = 1/|\mathbf{A}|$.

(iv) (Leibniz) $|\mathbf{A}| = \sum_\sigma \operatorname{sgn}(\sigma)\prod_i a_{i\sigma(i)}$; hence $|\mathbf{A}|$ is a polynomial in the entries, continuous and infinitely differentiable, and $\{|\mathbf{A}| \neq 0\}$ is open. $|\mathbf{I}_p| = 1$, $|\operatorname{diag}(d_i)| = \prod d_i$, $|c\mathbf{A}| = c^p|\mathbf{A}|$.

(v) (Adjugate) If $|\mathbf{A}| \neq 0$, $\mathbf{A}^{-1} = \operatorname{adj}(\mathbf{A})/|\mathbf{A}|$, so $\mathbf{A} \mapsto \mathbf{A}^{-1}$ is continuous and differentiable on $\{|\mathbf{A}| \neq 0\}$.

(vi) (Cauchy–Schwarz) $|\sum_k a_k b_k| \le (\sum a_k^2)^{1/2}(\sum b_k^2)^{1/2}$.

Definition 2.2: Frobenius norm

$\|\mathbf{A}\|_F = (\sum_{i,j} a_{ij}^2)^{1/2}$ and $\|\mathbf{x}\| = \|\mathbf{x}\|_F$ for vectors. Example: $\mathbf{M} = \begin{pmatrix} -0.1 & 0.3 \\ 0.2 & -0.2 \end{pmatrix}$ has $\|\mathbf{M}\|_F = \sqrt{0.18} \approx 0.424$. Estimation errors of the sample moments can be measured by $E_{\bar u} = \|\bar{\mathbf{u}} - \boldsymbol{\mu}\|_2$, $E_S = \|\mathbf{S} - \boldsymbol{\Sigma}\|_F$, $E_R = \|\mathbf{R} - \mathbf{R}_0\|_F$.

Theorem 2.3: basic identities

For $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{y} \in \mathbb{R}^q$, $\mathbf{A} \in \mathbb{R}^{p\times q}$, $\mathbf{B} \in \mathbb{R}^{q\times p}$:

(i) $\mathbf{x}^\top\mathbf{A}\mathbf{y} = \sum_{i,j} a_{ij}x_i y_j$; if $\mathbf{A}$ is diagonal, $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_i a_{ii}x_i^2$.

(ii) $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$.

(iii) $\|\mathbf{A}\|_F^2 = \operatorname{tr}(\mathbf{A}^\top\mathbf{A})$ and $\|\mathbf{x}\|^2 = \mathbf{x}^\top\mathbf{x}$.

Proof. (i) $(\mathbf{A}\mathbf{y})_i = \sum_j a_{ij}y_j$, so $\mathbf{x}^\top\mathbf{A}\mathbf{y} = \sum_i x_i(\mathbf{A}\mathbf{y})_i$.
(ii) $\operatorname{tr}(\mathbf{AB}) = \sum_i\sum_j a_{ij}b_{ji} = \sum_j\sum_i b_{ji}a_{ij} = \operatorname{tr}(\mathbf{BA})$.
(iii) $(\mathbf{A}^\top\mathbf{A})_{jj} = \sum_i a_{ij}^2$; sum over j. ■

Theorem 2.4: first-order expansion of the determinant

$|\mathbf{I}_p + t\mathbf{N}| = 1 + t\operatorname{tr}(\mathbf{N}) + O(t^2)$ as $t \to 0$.

Proof. In the Leibniz formula a non-identity σ moves at least two indices, so its term is $O(t^2)$. The identity contributes $\prod_i(1 + tn_{ii}) = 1 + t\sum_i n_{ii} + O(t^2)$. ■

Definition 2.5: open sets and continuity

With distance $\|\mathbf{B}-\mathbf{A}\|_F$: U is open iff every $\mathbf{A}\in U$ has a ball of radius $r>0$ inside U; f is continuous at $\mathbf{A}$ iff $\forall\epsilon\,\exists\delta$ with $\|\mathbf{B}-\mathbf{A}\|_F<\delta\Rightarrow\|f(\mathbf{B})-f(\mathbf{A})\|_F<\epsilon$. If every entry of a map is continuous, so is the map ($\epsilon/\sqrt{m}$ argument).

1.1 Block-matrix formulas

Used for conditional distributions, the Wishart theorem and likelihood ratio tests.

Theorem 2.6: block determinant and inverse

For $\mathbf{A}=\begin{pmatrix}\mathbf{A}_{11} & \mathbf{A}_{12}\\ \mathbf{A}_{21} & \mathbf{A}_{22}\end{pmatrix}$ with $\mathbf{A}_{22}$ invertible,

$$|\mathbf{A}|=|\mathbf{A}_{22}|\,|\mathbf{A}_{11}-\mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}|=|\mathbf{A}_{11}|\,|\mathbf{A}_{22}-\mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}|\quad(\mathbf{A}_{11}\text{ invertible}).$$

With the Schur complement $\mathbf{A}_{11\cdot2}:=\mathbf{A}_{11}-\mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$,

$$\mathbf{A}^{-1}=\begin{pmatrix}\mathbf{A}_{11\cdot2}^{-1} & -\mathbf{A}_{11\cdot2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1}\\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot2}^{-1} & \mathbf{A}_{22}^{-1}+\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1}\end{pmatrix};$$

in particular $(\mathbf{A}^{-1})_{11}=\mathbf{A}_{11\cdot2}^{-1}$.

Proof. Block Gaussian elimination:

$$\begin{pmatrix}\mathbf{I} & -\mathbf{A}_{12}\mathbf{A}_{22}^{-1}\\ \mathbf{0} & \mathbf{I}\end{pmatrix}\mathbf{A}\begin{pmatrix}\mathbf{I} & \mathbf{0}\\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{I}\end{pmatrix}=\begin{pmatrix}\mathbf{A}_{11\cdot2} & \mathbf{0}\\ \mathbf{0} & \mathbf{A}_{22}\end{pmatrix}.$$

The outer factors have determinant 1, giving the determinant formula; inverting both sides gives $\mathbf{A}^{-1}$. ■

Theorem 2.7: matrix determinant lemma and Woodbury identity

$$|\mathbf{A}+\mathbf{u}\mathbf{v}^\top|=|\mathbf{A}|(1+\mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}),\qquad|\mathbf{I}_p+\mathbf{u}\mathbf{v}^\top|=1+\mathbf{v}^\top\mathbf{u},$$
$$(\mathbf{A}+\mathbf{U}\mathbf{C}\mathbf{V})^{-1}=\mathbf{A}^{-1}-\mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1}+\mathbf{V}\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{A}^{-1}.$$

Proof. Compute $\begin{vmatrix}\mathbf{A} & -\mathbf{u}\\ \mathbf{v}^\top & 1\end{vmatrix}$ with both forms of the block formula: $|\mathbf{A}|(1+\mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u})=1\cdot|\mathbf{A}+\mathbf{u}\mathbf{v}^\top|$. For Woodbury, multiply the right side by $\mathbf{A}+\mathbf{U}\mathbf{C}\mathbf{V}$ and check that the product is $\mathbf{I}$. ■

2 Eigenvalues and Positive Definiteness

Motivation. Most multivariate statistics are functions of eigenvalues of covariance matrices; their basic properties are recorded here.

Theorem 2.8: spectral decomposition and consequences

Symmetric $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}(\mathbf{A})\mathbf{U}^\top = \sum_i \lambda_i(\mathbf{A})\mathbf{u}_i\mathbf{u}_i^\top$, $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_p(\mathbf{A})$ ($\lambda_i(\mathbf{A})$ denotes the ith largest eigenvalue).

$$|\mathbf{A}| = \prod_i \lambda_i(\mathbf{A}), \qquad \operatorname{tr}(\mathbf{A}) = \sum_i \lambda_i(\mathbf{A}), \qquad \mathbf{A} \succ 0 \iff \text{all } \lambda_i(\mathbf{A}) > 0.$$

Proof. $|\mathbf{U}||\mathbf{U}^\top| = |\mathbf{U}\mathbf{U}^\top| = 1$ gives $|\mathbf{A}| = |\mathbf{\Lambda}(\mathbf{A})|$; $\operatorname{tr}(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top) = \operatorname{tr}(\mathbf{\Lambda}\mathbf{U}^\top\mathbf{U}) = \operatorname{tr}\mathbf{\Lambda}$. Definiteness follows from $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_i \lambda_i(\mathbf{u}_i^\top\mathbf{x})^2$. ■

Theorem 2.9: smallest eigenvalue

For symmetric $\mathbf{A}, \mathbf{B}$:
(i) $\lambda_p(\mathbf{A}) = \min_{\|\mathbf{x}\|=1} \mathbf{x}^\top\mathbf{A}\mathbf{x}$;
(ii) $|\lambda_p(\mathbf{A}) - \lambda_p(\mathbf{B})| \leq \|\mathbf{A} - \mathbf{B}\|_F$;
(iii) $\{\mathbf{A} \succ 0\}$ is convex and open in the space of symmetric matrices.

Proof. (i) With $\mathbf{z} = \mathbf{U}^\top\mathbf{x}$, $\|\mathbf{z}\| = 1$ and $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum \lambda_i z_i^2 \geq \lambda_p$, equality at $\mathbf{x} = \mathbf{u}_p$.

(ii) For $\|\mathbf{x}\| = 1$, Cauchy–Schwarz gives $|\mathbf{x}^\top\mathbf{A}\mathbf{x} - \mathbf{x}^\top\mathbf{B}\mathbf{x}| = |\sum_{ij}(a_{ij} - b_{ij})x_i x_j| \leq \|\mathbf{A} - \mathbf{B}\|_F(\sum_{ij} x_i^2 x_j^2)^{1/2} = \|\mathbf{A} - \mathbf{B}\|_F$; minimize both sides over $\mathbf{x}$.

(iii) Convex: $\mathbf{x}^\top\{(1-s)\mathbf{A}_0 + s\mathbf{A}_1\}\mathbf{x} > 0$. Open: if $\mathbf{A} \succ 0$ and $\|\mathbf{B} - \mathbf{A}\|_F < \lambda_p(\mathbf{A})$, (ii) gives $\lambda_p(\mathbf{B}) \geq \lambda_p(\mathbf{A}) - \|\mathbf{A} - \mathbf{B}\|_F > 0$. ■

Theorem 2.10: symmetric square root

For $\mathbf{A} \succ 0$, $\mathbf{A}^{\pm 1/2} = \mathbf{U}\mathbf{\Lambda}^{\pm 1/2}\mathbf{U}^\top$ are symmetric, $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}$, $\mathbf{A}^{-1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1}$, $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}_p$. $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}^{-1}\mathbf{U}^\top \succ 0$ and $\lambda_i(\mathbf{A}^{-1}) = 1/\lambda_{p+1-i}(\mathbf{A})$ with eigenvector $\mathbf{u}_{p+1-i}$.

3 Eigenvalue Inequalities

Motivation. PCA, canonical correlation and discriminant analysis all maximize a ratio of quadratic forms. The extremal characterization of eigenvalues (Rayleigh quotient, Courant–Fischer) is the common tool that solves them.

Definition 2.11: Birkhoff's theorem and the rearrangement inequality

(i) A non-negative matrix with all row and column sums 1 (doubly stochastic) is a convex combination of permutation matrices.
(ii) If $a_1 \geq \cdots \geq a_p$ and $b_1 \geq \cdots \geq b_p$, then for every permutation σ, $\sum_i a_i b_{p+1-i} \leq \sum_i a_i b_{\sigma(i)} \leq \sum_i a_i b_i$.

Theorem 2.12: von Neumann's trace inequality

For symmetric $\mathbf{A} = \sum_i \lambda_i(\mathbf{A})\mathbf{u}_i\mathbf{u}_i^\top$, $\mathbf{B} = \sum_j \lambda_j(\mathbf{B})\mathbf{v}_j\mathbf{v}_j^\top$,

$$\sum_{i=1}^{p} \lambda_{p+1-i}(\mathbf{A})\lambda_i(\mathbf{B}) \leq \operatorname{tr}(\mathbf{A}\mathbf{B}) \leq \sum_{i=1}^{p} \lambda_i(\mathbf{A})\lambda_i(\mathbf{B}).$$

The upper bound is attained when $\mathbf{v}_i = \mathbf{u}_i$ and the lower bound when $\mathbf{v}_i = \mathbf{u}_{p+1-i}$.

Proof. Step 1. $\operatorname{tr}(\mathbf{AB}) = \sum_{i,j} \lambda_i(\mathbf{A})\lambda_j(\mathbf{B}) w_{ij}$, $w_{ij} = (\mathbf{u}_i^\top \mathbf{v}_j)^2 \geq 0$. Since $\{\mathbf{u}_i\}, \{\mathbf{v}_j\}$ are orthonormal bases, $\sum_j w_{ij} = \|\mathbf{u}_i\|^2 = 1$ and $\sum_i w_{ij} = \|\mathbf{v}_j\|^2 = 1$: $\mathbf{W}$ is doubly stochastic. **Step 2.** By Birkhoff $\mathbf{W}$ is a convex combination of permutation matrices; $\operatorname{tr}(\mathbf{AB})$ is linear in $\mathbf{W}$, so it lies between the minimum and maximum of $\sum_i \lambda_i(\mathbf{A})\lambda_{\sigma(i)}(\mathbf{B})$. **Step 3.** By rearrangement the maximum is at $\sigma(i) = i$, the minimum at $\sigma(i) = p+1-i$. **Step 4.** If $\mathbf{v}_i = \mathbf{u}_i$ (resp. $\mathbf{u}_{p+1-i}$), $\mathbf{W}$ is the identity (resp. reversal) permutation and the bound is attained. ■

Theorem 2.13: an elementary inequality

$x - \log x \geq 1$ for $x > 0$, with equality iff $x = 1$.

Proof. $\phi(x) = x - \log x$ has $\phi'(x) = 1 - 1/x$, negative on $(0,1)$, positive on $(1,\infty)$. For $x \neq 1$ the mean value theorem gives $\phi(x) - \phi(1) = \phi'(\xi)(x-1) > 0$ with ξ between 1 and x. ■

4 Vector Calculus

Motivation. Maximum likelihood for the multivariate normal requires derivatives with respect to vectors and matrices, including $\log\det \mathbf{\Sigma}$. These rules are collected here.

Definition 2.14: gradient, Jacobian, Hessian

For $f : \mathbb{R}^p \to \mathbb{R}$, $\partial f/\partial \mathbf{x} = (\partial f/\partial x_1, \ldots, \partial f/\partial x_p)^\top$. The Jacobian of $\mathbf{g} : \mathbb{R}^p \to \mathbb{R}^q$ is the $q \times p$ matrix $(\partial g_i/\partial x_j)$. The Hessian $\partial^2 f/\partial \mathbf{x}\partial \mathbf{x}^\top$ is the Jacobian of the gradient; it is symmetric when second partials are continuous (Schwarz's theorem).

Theorem 2.15: chain rule

For $\mathbf{y} = \mathbf{g}(\mathbf{x})$ with $q \times p$ Jacobian $\mathbf{J} = \partial \mathbf{g}/\partial \mathbf{x}^\top$,

$$\frac{\partial f(\mathbf{y})}{\partial \mathbf{x}} = \mathbf{J}^\top \frac{\partial f}{\partial \mathbf{y}}, \qquad \frac{\partial^2 f(\mathbf{y})}{\partial \mathbf{x}\partial \mathbf{x}^\top} = \mathbf{J}^\top \frac{\partial^2 f}{\partial \mathbf{y}\partial \mathbf{y}^\top}\mathbf{J} + \sum_{i=1}^{q} \frac{\partial f}{\partial y_i}\frac{\partial^2 g_i}{\partial \mathbf{x}\partial \mathbf{x}^\top}.$$

For an affine map $\mathbf{g}(\mathbf{x}) = \mathbf{B}\mathbf{x} + \mathbf{b}$, $\mathbf{J} = \mathbf{B}$ and the last sum vanishes.

Theorem 2.16: derivatives of quadratic forms and basic maps

For symmetric $\mathbf{A}$, $\dfrac{\partial\, \mathbf{x}^\top \mathbf{A}\mathbf{x}}{\partial \mathbf{x}} = 2\mathbf{A}\mathbf{x}$ and $\dfrac{\partial^2\, \mathbf{x}^\top \mathbf{A}\mathbf{x}}{\partial \mathbf{x}\partial \mathbf{x}^\top} = 2\mathbf{A}$. Also $\partial \mathbf{x}/\partial \mathbf{x}^\top = \mathbf{I}_p$, $\partial(\mathbf{B}\mathbf{x} + \mathbf{b})/\partial \mathbf{x}^\top = \mathbf{B}$, and $\partial(\mathbf{x} - \boldsymbol{\mu})/\partial \boldsymbol{\mu}^\top = -\mathbf{I}_p$.

Proof. $\mathbf{x}^\top \mathbf{A}\mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$, so $\partial/\partial x_k = \sum_j a_{kj} x_j + \sum_i a_{ik} x_i = 2(\mathbf{A}\mathbf{x})_k$; differentiating in x_l gives $2a_{kl}$. ■

5 Matrix Calculus

To optimize over covariance matrices we differentiate with respect to a matrix: matrix gradients give first-order conditions and directional derivatives give second-order conditions along symmetric

directions.

Definition 2.17: directional derivatives and matrix gradient

For f on an open set C, $\boldsymbol{\Theta} \in C$ and a direction $\mathbf{H}$,

$$Df(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d}{dt} f(\boldsymbol{\Theta} + t\mathbf{H})\Big|_{t=0}, \qquad D^2 f(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d^2}{dt^2} f(\boldsymbol{\Theta} + t\mathbf{H})\Big|_{t=0}.$$

The matrix gradient $\partial f / \partial \mathbf{X}$ has (i,j) entry $\partial f / \partial x_{ij}$.

Theorem 2.18: directional derivative and gradient

In $\mathbb{R}^p$: $Df(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top \frac{\partial f}{\partial \boldsymbol{\mu}}$, $D^2 f(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top \frac{\partial^2 f}{\partial \boldsymbol{\mu} \partial \boldsymbol{\mu}^\top} \mathbf{h}$. For matrices: $Df(\mathbf{X})[\mathbf{H}] = \operatorname{tr}\Big\{\Big(\frac{\partial f}{\partial \mathbf{X}}\Big)^\top \mathbf{H}\Big\} = \sum_{i,j} \frac{\partial f}{\partial x_{ij}} h_{ij}$.

Theorem 2.19: derivatives of basic matrix functions

For $\boldsymbol{\Sigma} \succ 0$, symmetric $\mathbf{H}$ and $p \times p$ $\mathbf{B}$:

$$D \log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}), \qquad D^2 \log|\boldsymbol{\Sigma}|[\mathbf{H}] = -\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H}),$$

$$D \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = \operatorname{tr}(\mathbf{H}\mathbf{B}), \qquad D^2 \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = 0,$$

$$\frac{\partial \log|\boldsymbol{\Sigma}|}{\partial \boldsymbol{\Sigma}} = \boldsymbol{\Sigma}^{-1}, \qquad \frac{\partial \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})}{\partial \boldsymbol{\Sigma}} = \mathbf{B}^\top.$$

D and D^2 are linear.

Proof. Let $\mathbf{M} = \boldsymbol{\Sigma}^{-1/2}\mathbf{H}\boldsymbol{\Sigma}^{-1/2} = \mathbf{V}\boldsymbol{\Lambda}(\mathbf{M})\mathbf{V}^\top$ (symmetric). Then $\boldsymbol{\Sigma} + t\mathbf{H} = \boldsymbol{\Sigma}^{1/2}(\mathbf{I} + t\mathbf{M})\boldsymbol{\Sigma}^{1/2}$ and $\mathbf{I} + t\mathbf{M}$ has eigenvalues $1 + t\lambda_k(\mathbf{M})$, so for small $|t|$

$$\log|\boldsymbol{\Sigma} + t\mathbf{H}| = \log|\boldsymbol{\Sigma}| + \sum_k \log\{1 + t\lambda_k(\mathbf{M})\}.$$

The first and second derivatives at $t = 0$ are $\sum_k \lambda_k(\mathbf{M}) = \operatorname{tr}\mathbf{M} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H})$ and $-\sum_k \lambda_k(\mathbf{M})^2 = -\operatorname{tr}(\mathbf{M}^2) = -\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H})$. $\operatorname{tr}\{(\boldsymbol{\Sigma} + t\mathbf{H})\mathbf{B}\}$ is linear in t. Gradients: for any $\mathbf{H}$, $|\boldsymbol{\Sigma} + t\mathbf{H}| = |\boldsymbol{\Sigma}||\mathbf{I} + t\boldsymbol{\Sigma}^{-1}\mathbf{H}| = |\boldsymbol{\Sigma}|\{1 + t\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}) + O(t^2)\}$, so $D \log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}\{(\boldsymbol{\Sigma}^{-1})^\top \mathbf{H}\}$; compare with Theorem 2.18. Likewise $\operatorname{tr}(\mathbf{H}\mathbf{B}) = \operatorname{tr}\{(\mathbf{B}^\top)^\top \mathbf{H}\}$. ■

Theorem 2.20: symmetric matrix variables

If $\mathbf{X}$ is symmetric and $\mathbf{G} = \partial f / \partial \mathbf{X}$ is symmetric, then $Df(\mathbf{X})[\mathbf{H}] = 0$ for all symmetric $\mathbf{H}$ iff $\mathbf{G} = \mathbf{0}$.

Proof. $\Leftarrow$ is clear. $\Rightarrow$: take $\mathbf{H} = \mathbf{G}$: $0 = \operatorname{tr}(\mathbf{G}^\top \mathbf{G}) = \|\mathbf{G}\|_F^2$. ■

6 Optimization

Motivation. Several estimators are defined as maximizers subject to constraints. Lagrange multipliers and the trace inequality used for the normal MLE are stated here.

Theorem 2.21: convexity and unique minimizer

Let C be open and convex and suppose $D^2 f(\boldsymbol{\Theta})[\mathbf{H}] > 0$ for all $\boldsymbol{\Theta} \in C$, $\mathbf{H} \neq \mathbf{0}$. Then (i) f is strictly convex on C;
(ii) a point $\hat{\boldsymbol{\Theta}}$ with $Df(\hat{\boldsymbol{\Theta}})[\mathbf{H}] = 0$ for all $\mathbf{H}$ is the unique global minimizer;
(iii) for $C = \mathbb{R}^p$ these read: Hessian $\succ 0$ and gradient $= \mathbf{0}$.

Proof. $g(t) = f(\boldsymbol{\Theta}_0 + t(\boldsymbol{\Theta}_1 - \boldsymbol{\Theta}_0))$ has $g''(t) = D^2 f(\cdot)[\mathbf{H}] > 0$, so g is strictly convex.

(ii) With $\boldsymbol{\Theta}_0 = \hat{\boldsymbol{\Theta}}$, $g'(0) = 0$ and g' is strictly increasing, so $g'(t) > 0$ on $(0, 1]$ and $f(\boldsymbol{\Theta}_1) = g(1) > g(0) = f(\hat{\boldsymbol{\Theta}})$ by the mean value theorem. ■

Theorem 2.22: transformations that keep the optimizer; profiling

For $a > 0$ and a constant c: $\arg\min\{af + c\} = \arg\min f$, $\arg\max f = \arg\min(-f)$, $\arg\max L = \arg\max \log L$ for $L > 0$, and for a bijection ϕ, $\arg\min_\theta f(\theta) = \phi^{-1}(\arg\min_\eta f \circ \phi^{-1}(\eta))$. **Profiling**: if $\hat{\theta}(\eta)$ minimizes $\ell(\cdot, \eta)$ for each η and $\hat{\eta}$ minimizes the profile $p(\eta) = \ell(\hat{\theta}(\eta), \eta)$, then $(\hat{\theta}(\hat{\eta}), \hat{\eta})$ minimizes ℓ jointly: $\ell(\theta, \eta) \geq \ell(\hat{\theta}(\eta), \eta) = p(\eta) \geq p(\hat{\eta})$.

The MLE (Ch. 4) first profiles out $\boldsymbol{\mu}$ and then minimizes over $\boldsymbol{\Sigma}$.

7 Probability and Convergence

Motivation. Large-sample results need convergence in probability and distribution for random vectors; the multivariate CLT and Slutsky's lemma are the tools.

Definition 2.23: standard results from mathematical statistics

(S0) $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$: $P(\|\mathbf{Y}_n - \mathbf{c}\| > \epsilon) \to 0$ for all $\epsilon > 0$. $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$: $\mathrm{E}g(\mathbf{Y}_n) \to \mathrm{E}g(\mathbf{Y})$ for all bounded continuous g.

(S1) Markov: for $W \geq 0$, $P(W \geq a) \leq \mathrm{E}W/a$.

(S2) $\mathrm{E}(\mathbf{AY} + \mathbf{b}) = \mathbf{A}\mathrm{E}\mathbf{Y} + \mathbf{b}$, $\mathrm{Cov}(\mathbf{AY} + \mathbf{b}) = \mathbf{A}\mathrm{Cov}(\mathbf{Y})\mathbf{A}^\top$; for independent $\mathbf{Y}_j$, $\mathrm{Cov}(\sum \mathbf{Y}_j) = \sum \mathrm{Cov}(\mathbf{Y}_j)$; $\mathrm{E}\|\mathbf{Y} - \mathrm{E}\mathbf{Y}\|^2 = \mathrm{tr}\mathrm{Cov}(\mathbf{Y})$.

(S3) Weak law of large numbers; (S4) univariate CLT.

(S5) Continuous mapping theorem (CMT): $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$, g continuous $\Rightarrow g(\mathbf{Y}_n) \xrightarrow{d} g(\mathbf{Y})$; $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$, g continuous at $\mathbf{c} \Rightarrow g(\mathbf{Y}_n) \xrightarrow{p} g(\mathbf{c})$.

(S6) Slutsky: $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$, $\mathbf{W}_n \xrightarrow{p} \mathbf{c} \Rightarrow (\mathbf{Y}_n, \mathbf{W}_n) \xrightarrow{d} (\mathbf{Y}, \mathbf{c})$.

(S7) Componentwise convergence in probability to constants implies joint convergence.

Theorem 2.24: vector Markov inequality and matrix convergence

$P(\|\mathbf{Y}\| > \epsilon) \leq \mathrm{E}\|\mathbf{Y}\|^2/\epsilon^2$. For random matrices, $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$ (entrywise) $\iff \|\mathbf{A}_n - \mathbf{A}\|_F \xrightarrow{p} 0$.

Proof. $\{\|\mathbf{Y}\| > \epsilon\} \subset \{\|\mathbf{Y}\|^2 \geq \epsilon^2\}$; apply Markov. Matrices: $\|\mathbf{A}_n - \mathbf{A}\|_F^2 \leq pq \max_{ij}(\mathbf{A}_n - \mathbf{A})_{ij}^2$, so $\{\|\mathbf{A}_n - \mathbf{A}\|_F > \epsilon\} \subset \bigcup_{ij}\{|(\mathbf{A}_n - \mathbf{A})_{ij}| > \epsilon/\sqrt{pq}\}$ and a union bound gives convergence; conversely $|(\mathbf{A}_n - \mathbf{A})_{ij}| \leq \|\mathbf{A}_n - \mathbf{A}\|_F$. ■

Theorem 2.25: characteristic functions and the Cramér–Wold device

$\varphi_{\mathbf{Y}}(\mathbf{t}) = \mathrm{E}\exp(i\mathbf{t}^\top\mathbf{Y})$.
(i) Uniqueness: $\varphi_{\mathbf{Y}} \equiv \varphi_{\mathbf{W}} \iff \mathbf{Y} \overset{d}{=} \mathbf{W}$.
(ii) Lévy continuity: $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y} \iff \varphi_{\mathbf{Y}_n}(\mathbf{t}) \to \varphi_{\mathbf{Y}}(\mathbf{t})$ for all $\mathbf{t}$. **Cramér–Wold**:
(i) the distribution of $\mathbf{Y}$ is determined by those of all $\mathbf{t}^\top\mathbf{Y}$;
(ii) $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y} \iff \mathbf{t}^\top\mathbf{Y}_n \xrightarrow{d} \mathbf{t}^\top\mathbf{Y}$ for every $\mathbf{t}$.

Proof. The key identity is $\varphi_{\mathbf{t}^\top\mathbf{Y}}(1) = \mathrm{E}e^{i\mathbf{t}^\top\mathbf{Y}} = \varphi_{\mathbf{Y}}(\mathbf{t})$.

(i) If $\mathbf{t}^\top\mathbf{Y} \overset{d}{=} \mathbf{t}^\top\mathbf{W}$, then $\varphi_{\mathbf{Y}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top\mathbf{Y}}(1) = \varphi_{\mathbf{t}^\top\mathbf{W}}(1) = \varphi_{\mathbf{W}}(\mathbf{t})$.

(ii) $\Leftarrow$: Lévy for scalars gives $\varphi_{\mathbf{t}^\top\mathbf{Y}_n}(s) \to \varphi_{\mathbf{t}^\top\mathbf{Y}}(s)$; take $s = 1$. ■

"All one-dimensional linear projections determine the multivariate distribution" underlies the general definition of the multivariate normal (Ch. 3) and the multivariate CLT.

8 Selected exercises

Motivation. Three further tools: the min–max characterization of every eigenvalue (which makes sample eigenvalues consistent), Hadamard's determinant inequality (which bounds $|\mathbf{R}|$ in independence tests), and the Kronecker product (the language for the covariance of a whole data matrix).

Exercise 2.1 (Courant–Fischer and Weyl)

Let $\mathbf{A}$ be symmetric $p\times p$ with eigenvalues $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_p(\mathbf{A})$ and orthonormal eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_p$.
(a) Show that for $k = 1, \ldots, p$,

$$\lambda_k(\mathbf{A}) = \max_{\mathcal{V}:\ \dim\mathcal{V}=k}\ \min_{\mathbf{x}\in\mathcal{V},\ \|\mathbf{x}\|=1} \mathbf{x}^\top\mathbf{A}\mathbf{x}.$$

(b) For symmetric $\mathbf{A}, \mathbf{B}$ show $|\lambda_k(\mathbf{A}) - \lambda_k(\mathbf{B})| \leq \|\mathbf{A}-\mathbf{B}\|_F$ for every k, which extends Theorem 2.9(ii) to all eigenvalues.

Solution [Supplement].

(a)

Step 1 ($\geq$). Let $\mathcal{V}_k = \mathrm{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$. A unit vector $\mathbf{x} \in \mathcal{V}_k$ is $\mathbf{x} = \sum_{i\leq k} c_i\mathbf{u}_i$ with $\sum_i c_i^2 = 1$, so $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_{i\leq k}\lambda_i c_i^2 \geq \lambda_k$. Hence the minimum over $\mathcal{V}_k$ is at least λ_k (it equals λ_k at $\mathbf{x} = \mathbf{u}_k$), and the maximum over all $\mathcal{V}$ is at least λ_k.

Step 2 ($\leq$). Let $\mathcal{V}$ be any subspace with $\dim\mathcal{V} = k$ and $\mathcal{W} = \mathrm{span}(\mathbf{u}_k, \ldots, \mathbf{u}_p)$, $\dim\mathcal{W} = p-k+1$. Since $\dim\mathcal{V} + \dim\mathcal{W} = p+1 > p$, the intersection $\mathcal{V}\cap\mathcal{W}$ contains a unit vector $\mathbf{x} = \sum_{i\geq k} c_i\mathbf{u}_i$, and $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_{i\geq k}\lambda_i c_i^2 \leq \lambda_k$. So the minimum over $\mathcal{V}$ is at most λ_k for every $\mathcal{V}$.

(b)

Step 1. Let $\epsilon = \|\mathbf{A}-\mathbf{B}\|_F$. As in the proof of Theorem 2.9(ii), $|\mathbf{x}^\top\mathbf{A}\mathbf{x} - \mathbf{x}^\top\mathbf{B}\mathbf{x}| \leq \epsilon$ for every unit $\mathbf{x}$.

Step 2. Fix $\mathcal{V}$ with $\dim\mathcal{V} = k$. The unit sphere of $\mathcal{V}$ is compact, so $\mathbf{x} \mapsto \mathbf{x}^\top\mathbf{B}\mathbf{x}$ attains its minimum on it at some $\mathbf{x}_0$. Then

$$\min_{\mathbf{x}\in\mathcal{V},\|\mathbf{x}\|=1} \mathbf{x}^\top\mathbf{A}\mathbf{x} \leq \mathbf{x}_0^\top\mathbf{A}\mathbf{x}_0 \leq \mathbf{x}_0^\top\mathbf{B}\mathbf{x}_0 + \epsilon = \min_{\mathbf{x}\in\mathcal{V},\|\mathbf{x}\|=1} \mathbf{x}^\top\mathbf{B}\mathbf{x} + \epsilon.$$

Step 3. Taking the maximum over $\mathcal{V}$ and using (a) gives $\lambda_k(\mathbf{A}) \le \lambda_k(\mathbf{B}) + \epsilon$. Exchanging $\mathbf{A}$ and $\mathbf{B}$ gives $\lambda_k(\mathbf{B}) \le \lambda_k(\mathbf{A}) + \epsilon$.

Consequently each $\lambda_k(\cdot)$ is continuous, and $\mathbf{S} \xrightarrow{p} \mathbf{\Sigma}$ implies $\lambda_k(\mathbf{S}) \xrightarrow{p} \lambda_k(\mathbf{\Sigma})$ for every k (sample PC variances are consistent). ■

Exercise 2.2 (Hadamard's inequality)

Let $\mathbf{A} \succ 0$ be $p \times p$. Show that $|\mathbf{A}| \le \prod_{i=1}^{p} a_{ii}$, with equality iff $\mathbf{A}$ is diagonal. Deduce that a correlation matrix $\mathbf{R} \succ 0$ satisfies $0 < |\mathbf{R}| \le 1$, with $|\mathbf{R}| = 1$ iff $\mathbf{R} = \mathbf{I}_p$.

Solution [Supplement].

Step 1 (reduction). $a_{ii} = \mathbf{e}_i^\top \mathbf{A} \mathbf{e}_i > 0$. Let $\mathbf{D} = \operatorname{diag}(a_{11}, \dots, a_{pp})$ and $\mathbf{B} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$. Then $\mathbf{B} \succ 0$, $b_{ii} = 1$ and $|\mathbf{B}| = |\mathbf{A}| / \prod_i a_{ii}$, so it suffices to show $|\mathbf{B}| \le 1$ with equality iff $\mathbf{B} = \mathbf{I}_p$.

Step 2. Let $\mu_1, \dots, \mu_p > 0$ be the eigenvalues of $\mathbf{B}$. By Theorem 2.8, $\sum_i \mu_i = \operatorname{tr}\mathbf{B} = p$ and $\log |\mathbf{B}| = \sum_i \log \mu_i$. Theorem 2.13 gives $\log \mu_i \le \mu_i - 1$, so

$$\log |\mathbf{B}| \le \sum_{i=1}^{p} (\mu_i - 1) = \operatorname{tr}\mathbf{B} - p = 0.$$

Step 3 (equality). Equality holds iff $\mu_i = 1$ for all i, i.e. $\mathbf{B} = \mathbf{U} \mathbf{I}_p \mathbf{U}^\top = \mathbf{I}_p$, i.e. $\mathbf{A} = \mathbf{D}$ is diagonal.

For a correlation matrix $a_{ii} = 1$, and $|\mathbf{R}| > 0$ because $\mathbf{R} \succ 0$. ■

Exercise 2.3 (Kronecker products and vec)

For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{r \times s}$ let $\mathbf{A} \otimes \mathbf{B} = [a_{ij}\mathbf{B}]_{i,j} \in \mathbb{R}^{mr \times ns}$, and let $\operatorname{vec}(\mathbf{X})$ stack the columns of $\mathbf{X}$.
(a) Show $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$ whenever the products are defined, $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$, and $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$ for invertible $\mathbf{A}, \mathbf{B}$.
(b) For symmetric $\mathbf{A}$ $(m \times m)$ and $\mathbf{B}$ $(r \times r)$ with eigenpairs $(\lambda_i, \mathbf{u}_i)$ and $(\mu_j, \mathbf{v}_j)$, show that $\mathbf{A} \otimes \mathbf{B}$ has the orthonormal eigenvectors $\mathbf{u}_i \otimes \mathbf{v}_j$ with eigenvalues $\lambda_i \mu_j$; hence $|\mathbf{A} \otimes \mathbf{B}| = |\mathbf{A}|^r |\mathbf{B}|^m$ and $\operatorname{tr}(\mathbf{A} \otimes \mathbf{B}) = \operatorname{tr}\mathbf{A} \operatorname{tr}\mathbf{B}$.
(c) Show $\operatorname{vec}(\mathbf{AXB}) = (\mathbf{B}^\top \otimes \mathbf{A}) \operatorname{vec}(\mathbf{X})$.
(d) If the $n \times p$ data matrix $\mathbf{X}$ has iid rows $N_p(\boldsymbol{\mu}, \mathbf{\Sigma})$, show $\operatorname{vec}(\mathbf{X}) \sim N_{np}(\boldsymbol{\mu} \otimes \mathbf{1}_n,\ \mathbf{\Sigma} \otimes \mathbf{I}_n)$.

Solution [Supplement].

(a) The (i,k) block of the product is $\sum_j (a_{ij}\mathbf{B})(c_{jk}\mathbf{D}) = (\sum_j a_{ij} c_{jk})\mathbf{BD} = (\mathbf{AC})_{ik}\mathbf{BD}$, the (i,k) block of $\mathbf{AC} \otimes \mathbf{BD}$. The (i,j) block of $(\mathbf{A} \otimes \mathbf{B})^\top$ is the transpose of the (j,i) block, $a_{ji}\mathbf{B}^\top$. Finally $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^{-1} \otimes \mathbf{B}^{-1}) = \mathbf{I}_m \otimes \mathbf{I}_r = \mathbf{I}_{mr}$.

(b)

Step 1. Treating vectors as one-column matrices, (a) gives $(\mathbf{A} \otimes \mathbf{B})(\mathbf{u}_i \otimes \mathbf{v}_j) = \mathbf{A}\mathbf{u}_i \otimes \mathbf{B}\mathbf{v}_j = \lambda_i \mu_j\, \mathbf{u}_i \otimes \mathbf{v}_j$.

Step 2. $(\mathbf{u}_i \otimes \mathbf{v}_j)^\top (\mathbf{u}_k \otimes \mathbf{v}_l) = (\mathbf{u}_i^\top \mathbf{u}_k)(\mathbf{v}_j^\top \mathbf{v}_l) = \delta_{ik}\delta_{jl}$, so these mr vectors form an orthonormal eigenbasis of the symmetric matrix $\mathbf{A} \otimes \mathbf{B}$.

Step 3. By Theorem 2.8, $|\mathbf{A} \otimes \mathbf{B}| = \prod_{i,j} \lambda_i \mu_j = (\prod_i \lambda_i)^r (\prod_j \mu_j)^m = |\mathbf{A}|^r |\mathbf{B}|^m$ and $\operatorname{tr}(\mathbf{A} \otimes \mathbf{B}) = \sum_{i,j} \lambda_i \mu_j = \operatorname{tr}\mathbf{A} \operatorname{tr}\mathbf{B}$.

(c) Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have columns $\mathbf{x}_1, \dots, \mathbf{x}_p$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$. Column k of $\mathbf{AXB}$ is $\mathbf{AX}\mathbf{b}_k = \sum_j b_{jk}\mathbf{A}\mathbf{x}_j = [b_{1k}\mathbf{A}, \dots, b_{pk}\mathbf{A}] \operatorname{vec}(\mathbf{X})$, and $[b_{1k}\mathbf{A}, \dots, b_{pk}\mathbf{A}]$ is the kth block row of $\mathbf{B}^\top \otimes \mathbf{A}$.

(d) Column j of $\mathbf{X}$ is $\mathbf{x}_{(j)} = (X_{1j}, \dots, X_{nj})^\top$ with mean $\mu_j \mathbf{1}_n$, so $\mathrm{E} \operatorname{vec}(\mathbf{X}) = \boldsymbol{\mu} \otimes \mathbf{1}_n$. For columns j, k, $\operatorname{Cov}(X_{ij}, X_{lk}) = \sigma_{jk}\delta_{il}$ by independence of the rows, so $\operatorname{Cov}(\mathbf{x}_{(j)}, \mathbf{x}_{(k)}) = \sigma_{jk}\mathbf{I}_n$, the (j,k) block

of $\boldsymbol{\Sigma} \otimes \mathbf{I}_n$. Normality: the stacked rows $(\mathbf{X}_1^\top, \ldots, \mathbf{X}_n^\top)^\top$ are jointly normal (independent normal blocks), and $\mathrm{vec}(\mathbf{X})$ is a permutation, hence a linear map, of this vector (Theorem 3.7). ■